Year 11 Foundation

REVISION PACK

Algebra 7

Name: _____

Class: _____

Algebra 7 · Formulae (Lessons 2.1, 2.2)

Worked example: substitution into P = 2l + 2w

$P = 2l + 2w$. Work out the value of P when $l = 9$ and $w = 5$

Step 1	Put the numbers in	$P = 2 \times 9 + 2 \times 5$
Step 2	Multiply first	$= 18 + 10$
Step 3	Answer	$P = 28$

Practice: substitution into P = 2l + 2w

1.1 $P = 2l + 2w$. Work out the value of P when $l = 8$ and $w = 5$

1.2 $P = 2l + 2w$. Work out the value of P when $l = 12$ and $w = 7$

1.3 $P = 2l + 2w$. Work out the value of P when $l = 3.5$ and $w = 2$

1.4 $P = 2l + 2w$. Work out the value of P when $l = 10$ and $w = 4.5$

1.5 $P = 2l + 2w$. Work out the value of P when $l = 15$ and $w = 9$

1.6 $P = 2l + 2w$. Work out the value of P when $l = 6$ and $w = 2.5$

Algebra 7 · Formulae (Lessons 2.1, 2.2)

Worked example: substitution into v = u + at

$v = u + at$. Work out the value of v when $u = 8$, $a = 2$ and $t = 6$

Step 1	Put the numbers in	$v = 8 + 2 \times 6$
Step 2	Multiply first	$= 8 + 12$
Step 3	Answer	$v = 20$

Practice: substitution into v = u + at

1.7 $v = u + at$. Work out the value of v when $u = 10$, $a = 4$ and $t = 3$

1.8 $v = u + at$. Work out the value of v when $u = 5$, $a = 6$ and $t = 7$

1.9 $v = u + at$. Work out the value of v when $u = 15$, $a = 2$ and $t = 9$

1.10 $v = u + at$. Work out the value of v when $u = 3$, $a = 5$ and $t = 8$

1.11 $v = u + at$. Work out the value of v when $u = 20$, $a = 3$ and $t = 4$

1.12 $v = u + at$. Work out the value of v when $u = 7$, $a = 9$ and $t = 2$

Algebra 7 · Formulae (Lessons 2.1, 2.2)

Worked example: using a formula

The cost, T KD, of hiring a car for n hours is $T = 30n + 20$. Mona pays 140 KD. For how many hours did Mona hire the car?

Step 1	Make an equation	$30n + 20 = 140$
Step 2	Subtract 20	$30n = 120$
Step 3	Divide by 30	$n = 4$
Step 4	Answer	4 hours

Practice: using a formula

1.13 The cost, T KD, of hiring a car for n hours is $T = 35n + 15$. Ali pays 155 KD. For how many hours did Ali hire the car?

1.14 The cost, T KD, of hiring a bike for n hours is $T = 25n + 10$. Omar pays 135 KD. For how many hours did Omar hire the bike?

1.15 The cost, T KD, of hiring a boat for n hours is $T = 50n + 30$. Fahad pays 280 KD. For how many hours did Fahad hire the boat?

1.16 The cost, T KD, of hiring a van for n hours is $T = 45n + 20$. Sara pays 200 KD. For how many hours did Sara hire the van?

1.17 The cost, T KD, of hiring a jet ski for n hours is $T = 60n + 40$. Laila pays 340 KD. For how many hours did Laila hire the jet ski?

1.18 The cost, T KD, of hiring a kayak for n hours is $T = 20n + 15$. Yousef pays 135 KD. For how many hours did Yousef hire the kayak?

Worked example: rearranging formulae

Rearrange each formula to make the given letter the subject.

(a) $y = 3x + 5$ — Make x the subject.

(b) $A = \pi r l$ — Make l the subject.

(a)	**Step 1**	Subtract 5	$y - 5 = 3x$
	Step 2	Divide by 3	$x = \dfrac{y-5}{3}$
(b)	**Step 1**	Divide by πr	$l = \dfrac{A}{\pi r}$

Practice: rearranging formulae

1.19 Rearrange each formula to make the given letter the subject.

(a) $y = 5x + 2$ — Make x the subject.

(b) $A = lw$ — Make w the subject.

1.20 Rearrange each formula to make the given letter the subject.

(a) $y = 2x - 7$ — Make x the subject.

(b) $V = Ah$ — Make h the subject.

1.21 Rearrange each formula to make the given letter the subject.

(a) $y = 6x + 1$ — Make x the subject.

(b) $C = \pi d$ — Make d the subject.

1.22 Rearrange each formula to make the given letter the subject.

(a) $y = 3x - 4$ — Make x the subject.

(b) $P = 4s$ — Make s the subject.

1.23 Rearrange each formula to make the given letter the subject.

(a) $y = 8x + 3$ — Make x the subject.

(b) $F = ma$ — Make a the subject.

1.24 Rearrange each formula to make the given letter the subject.

(a) $y = 4x - 9$ — Make x the subject.

(b) $D = st$ — Make t the subject.